Background subtraction for diffraction frames: state of the problem

Subproject **pod/bgsub** (GitHub mit-eremin/pod-subtract, main at 00f13bf), Dmitry Eremin, Nelson Lab, Caltech. Split out of pod/background on 2026-11-05 at background commit f1e93f5; package **bgsub**, 47 tests pass, self-contained (vendored count law, detector, masks, centre, provenance; a frozen simulator-v1 frame bank, 716 frames reproduced bit for bit). Page 1 states the problem and where the algorithm stands by its own rules; page 2 measures it against pyREyes and XDS / superxds on real frames (compare v0, same day), then the defects and next steps. Session ids are anonymous (sids).

1. The problem, in the owner's words (re-scope 2026-11-04, scope correction 2026-11-05)

A very **fast**, **mathematically correct**, **universal** background estimation and subtraction algorithm for electron-diffraction frames, better than the flat local backgrounds of XDS / DIALS (flat boxes are biased on a curved background and lose intensity; their code is not copied). Correct means: unbiased background **everywhere** and **under masked spots**; honest per-pixel background **variance**; right behaviour under the detector's real **count statistics**; right **near the beam stop, grid bars and crystals**; and a fast estimate (target <= 50 ms per 1024^2 frame).

- **Background only.** Peak integration is not the product and the result is not for XDS. The integrators in the code (G-ML, LC, intervals, coverage) were the orchestrator's addition; they stay but nothing is judged on them. Injected peaks remain a *test*, scored against an oracle-background subtraction, never against true intensity.
- **No other packages** (no DIALS, nXDS); methods are developed here. The physics forward model / simulator stays in pod/background and is a test bench only.
- **Universal** today means Apollo (bin 4, 8, 16) with its stored-count law; CETA-D and TVIPS need their own variance models before any claim.

2. The approach (v1 default; v2, v2.1 opt-in; v0 reproducible)

The frame's stored counts are reduced to sufficient statistics on polar cells (1-px rings x 64 sectors x 2 checkerboard halves) and fitted with a **linear smooth model in stored counts**: a fine radial cubic B-spline (rings, steep halo) + azimuthal harmonics m <= 6 on a coarse radial spline + centre / ellipticity derivative columns + (v1) penumbra columns on the 1-4 px next to the beam stop; (v2) re-centring and a shrunken 1-px ring refinement, (v2.1) on elliptical rings. The fit is **cross-fitted weighted least squares**: the weights are the count-law variances taken from the *other* checkerboard half, so the estimator is linear in the counts and unbiased for *any* count law whenever the truth lies in the model's span (no median, no Poisson assumption). Spots, hot pixels and zingers are excluded by one-sided count-law tests (alpha 10^{-6} per pixel and per 3 x 3 window); spots the caller will measure are always excluded by discs. Variance is a sandwich with the frame's Pearson dispersion. The hybrid **H2.1** adds a local annulus residual only where its signal is significant (T > 2 with block standard errors), with the spot's own light removed (spill correction); otherwise it is the global **G2.1**.

The Apollo count law it carries: stored value = round-half-even of the 3-raw-frame electron sum / m (m = 1, 4, 16 e per count at bin 4, 8, 16), tie rule p = 0.403, paralyzable dead time tau = 0.206 per e per physical pixel per raw frame; archive TIFFs are MRC + 1; seams and borders masked.

3. Where it stands: v2.1 against its own pre-registered rules (2026-11-05)

Every version is pre-registered (rules committed before the test run, no parameter changes after it, failures reported with their cells). The verdict rule: VALID iff R1, R2, R3 and R5 pass and R4 passes in the F and classic regimes. **v2.1 is NOT VALID by that rule, by two narrow margins.** All thresholds (0.5 %, 3.2 SE, 50 ms, ...) were set by the orchestrator, not the owner; the owner ruled on 2026-11-05 that the **speed gate is relative** (single-CPU Python), so R5 is reported, not a verdict.

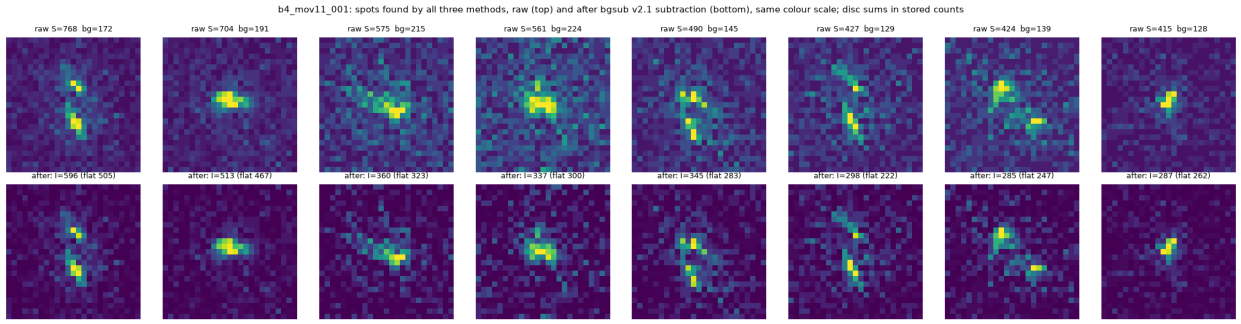
Rule	What it tests	v2 (previous)	v2.1 (current)	v2.1 status
R1	Synthetic unbiasedness: 60 lacey cells (bin 4 / 8 x dose x1 / 3 / 10 x plain / structured x D0-D4), |beta| <= max(0.5 %, 3.2 SE), for G and H	G pass 0/60; H fail 2/60 (arm cell +0.70 %, sparse cell -3.16 %)	G2.1 pass 0/60; H2.1 fail 1/60: sparse bin-8 cell (dose x3, plain, D1; 0.2 stored counts per box) -2.46 +- 0.72 %; arm cell fixed (+0.34 %)	FAIL 1/60 (H); no power at 0.2 counts/box
R2	Count statistics / truncation: spot-free twins, 24 cells, paired bias of G <= max(0.3 %, 3.1 SE); masking on - off <= 0.1 %	pass (max 0.45 %)	pass (max 0.45 %)	PASS
R3	Sigma honesty and interval coverage of the injected-spot estimates (integration; out of scope now)	pass; coverage 0.57-0.58	pass; coverage 0.66-0.69 (fix 3)	PASS (not a product criterion)
R4	Real frames, better than FLAT per regime x d bin, spots without host overlap: F, BA, CN, classic (17 fly-scan + 20 classic test sessions, 48,930 spots; 9 confirmation sessions)	P / P / P / P	P / P / P / P; confirmation P; all-spot classic F (host Bragg overlap, not background)	PASS
R5	Speed of the background estimate: 1024^2, numba, plan cached, <= 50 ms on this Mac	pass 46.8 ms (own run), 50.2 ms in the v2.1 run	54.2 ms under load (v1 46.3 ms in the same run); elliptical ring kernels +2.7 ms (A278)	FAIL 54.2 ms

R6 and R7 (sharp ring) are reported, not in the verdict. Report of record: notes/reports/2026-11-05_bgsub_algorithm_v2_1.md (Sec. 2.4 rules, Sec. 7 verdict, Sec. 8 limits).

4. Measured against pyREyes and XDS / superxds on real frames (compare v0, 2026-11-05)

40 real Apollo frames: 30 bin-8 fly-scan snapshots (1024^2, 20 REyes Good + 10 Bad, two sessions, ~0.08 counts / px background) and 10 frames of a bin-4 rotation movie (2048^2, XDS's own SPOT.XDS as a reference). **pyREyes** 3.6.3 has no background model (difference of Gaussians, global mean + k std threshold, components, a count; no per-peak intensity; re-implemented and verified bit for bit). **superxds** is a line-by-line Fortran port of XDS (static box-mean table x one scalar per frame; Poisson strong-pixel test; 'intensity' = sum of significance labels); it does not build on macOS / arm64, so the installed XDS stood in. All peaks were measured on the ORIGINAL frame in the same disc (3 px at bin 8, 5 px at bin 4).

Question	bgsub v2.1	pyREyes	XDS / superxds	flat annulus (XDS / DIALS-style local box)
Background on spot-free pixels, by radius	unbiased: |residual| < 1 % in every 32-px band, both binnings; bin 8: mean and count-law variance match the data to 1e-4	none (band-pass only)	bin 4: -9 to -14 % in the halo, high in the tail; bin 8: collapses (0.07 % of pixels pass its background screen)	fit to the same pixels: not a test
Peaks per frame (median)	bin 8 Good 23 / Bad 14; bin 4 187 (count-law flags, >= 5 sigma per component)	bin 8 Good 14 / Bad 17 (half are single stray electrons; fires on the stop edge); bin 4 45	bin 8: 4 (defaults, 15-frame pseudo-sweep; single frames fail); bin 4: 46 (defaults), 9 (owner 4/7/5)	-
Agreement	holds 74-87 % of XDS and pyREyes peaks at bin 4 (74 % of XDS at bin 8) + 2-4x more weak high-angle candidates	87 % of its bin-4 peaks are in bgsub; 58 % in XDS	76 % in bgsub; 56 % in pyREyes	-
Background under the same peaks (median ratio to bgsub) and net intensity	1 (reference); net I median 23 (bin 8), 101 (bin 4) counts	no background	1.04 (bin 8), 1.01 (bin 4): agrees under peaks although wrong between them	**1.15 at both binnings: loses 5 % (bin 8) and 14 % (bin 4) of the net intensity, ~17 % on bright spots**
Time per frame (this Mac)	62 ms (1024^2), 155 ms (2048^2), Python + numba	31 / 149 ms (two blurs)	33 / 125 ms (Fortran, 4 threads)	-



Eight spots on one bin-4 frame found by all three methods: raw (top) and after bgsub subtraction (bottom), same colour scale; captions give the disc sum, bgsub's background and the net intensity, with the flat-annulus net in brackets (8-25 % lower on every spot). Full tables and the overlays of every method's peaks on the original frames: notes/reports/2026-11-05_compare_v0.md, notes/figures/compare_v0/.

5. Known defects and limits

- **Sparse counts (R1).** The one failing synthetic cell (0.2 counts per box) had no statistical power (SE 0.72 % vs a 0.5 % limit); on 64 fresh frames H2.1 is -0.47 +- 0.72 %. Needs a power-adequate re-test, possibly a count-level-aware gate.
- **Sharp rings (A277).** A film ring narrower than the spline knots (4.6 A, one classic session) is not removed: most of its pixels are count-law flags. Open option: a ring-zone high-breakdown first pass.
- **Coverage.** Apollo bin 8 and bin 4 tested on real frames; bin 16 synthetic only (variance not honest, A196); **CETA-D bin 2 and TVIPS F30 untested**, need an integrating-detector variance model first. Not modelled: structure neither smooth on the model's scales nor linear at the annulus scale (A279). compare v0 is descriptive, not pre-registered, no truth for real peaks.

6. Next steps (ROADMAP.md) and what the owner must decide

- **1. Background-only validity spec**, pre-registered, with thresholds the owner endorses: bias everywhere and under masked spots with stated power; per-pixel variance honesty; stop / bar / crystal zones; count-law regimes; timing reported, not gated. **2.** Bring v2.1 to valid under it (sparse-cell re-test; optional ring-zone pass). **3-5.** Integrating-detector variance model, then CETA-D bin 2 and TVIPS F30 validation; honest variance at bin 16.
- **Owner decisions:** (a) per-frame **output format** (proposed: float32 background map + per-pixel variance map in stored counts, subtracted frame on request, sidecar with version / count law / zero level / mask / centre); (b) **datasets** for CETA-D bin 2 and TVIPS F30; (c) whether to run superxds itself on a Linux x86-64 box (B1); (d) the mov11 XDS.INP beam centre is 75 px off and was never refined by XDS (B2): re-check the XDS-merge study.